

Théo Verhelst – CV

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Links [Personal website](#) – [LinkedIn](#) – [Scholar](#)
[ResearchGate](#) – [GitHub](#)



Summary

Researcher in causal machine learning with a PhD in computer science. My research ranges from early contributions to causal inference for customer churn prediction to my current role at the European Space Agency, where I design causal methods for space applications and investigate foundational questions in physics, always through the lens of causality. With a strong experience in modelling, programming and collaborative work, I am eager to translate this into clear, engaging teaching and mentorship at the professorial level.

Employment history

2024 – 2026

Research Fellow at the European Space Agency, [Advanced Concepts Team](#).

Research topic: causality for space applications and fundamental physics.

2020 – 2024

PhD student at the Machine Learning Group (project [Machu-Picchu](#)), in collaboration with Orange Belgium, funded by Innoviris.

Thesis: *Causal and predictive modelling of customer churn: lessons learned from empirical and theoretical research.*

2019 – 2020

Researcher at the Machine Learning Group (project [DefeatFraud](#)), in collaboration with Atos / Worldline, funded by Innoviris.

Research topic: transfer learning for credit card fraud detection.

2018

Back-end programmer at Open Summer of Code 2018.

Project: Development of a visualisation tool for Dutch open parking data.

Education

2020 – 2024

PhD in Computer Science at the Machine Learning Group, Université Libre de Bruxelles (ULB), in collaboration with Orange Belgium

PhD thesis: *Causal and predictive modelling of customer churn: lessons learned from empirical and theoretical research* [\[PDF\]](#)

Supervisor: Prof. Gianluca Bontempi

Defended in January 2024

2018 – 2019

Second year of Master in Computer Science at ULB

Master's thesis: *Churn Prediction and Causal Analysis on Telecom Customer Data* [\[PDF\]](#)

Supervisor: Prof. Gianluca Bontempi

Raymond Devillers award for best master thesis

Average mark: 18.2 / 20, Grande distinction

2017 – 2018

First year of Master in AI at the University of Southampton, Erasmus Programme exchange

Average mark: 16.4 / 20

2014 – 2017

Bachelor in Computer Science at ULB

Bachelor's thesis: *Pitch Shifting of Music Signals*

Average mark: 15.65 / 20, with Grande distinction

Research funding

- **Innoviris Applied PhD** grant (€370k, 4 years) co-written and awarded for a joint ULB–Orange Belgium project on causal machine learning for customer churn prevention.

Presentations

- 9th European Conference on Space Debris, 1st to 4th of April 2025, Bonn, Germany.
- ECML PKDD 2023 - Workshop on Uplift Modeling and Causal Machine Learning for Operational Decision Making, 18th to 22nd of September 2023, Turin, Italy.
- Fundamental Challenges in Causality, 9th to 12th of May 2023, Grenoble, France (poster presentation).
- 24th International Conference on Discovery Science (DS 2021), 11th to 13th of October 2021, Halifax, Canada.
- 31st Benelux Conference on Artificial Intelligence (BNAIC 2019) and the 28th Belgian Dutch Conference on Machine Learning (Benelearn 2019), 6th to 8th of November 2019, Brussels, Belgium.

Teaching activities

Academic years 2019-2020, 2021-2022, 2022-2023

TA for *Big Data: Distributed Data Management and Scalable Analytics* at ULB [\[INFO-H515\]](#)

Academic year 2020-2021

TA for *Statistical Foundations of Machine Learning* at ULB [[INFO-F422](#)]

Supervision

- Supervised a Master's thesis on machine learning for uplift modelling (2021-2022).
- Supervised an internship on building a database of simulated satellite collisions for an online prediction challenge (2025).

Service

- **Scientific committee** of SPAICE 2026, the third edition of the European Space Agency's AI conference in and for space.

Languages

French First language

English Proficient (CEFR C2)